

Simulation and Modeling

Lecture 11

Modeling Complex Systems

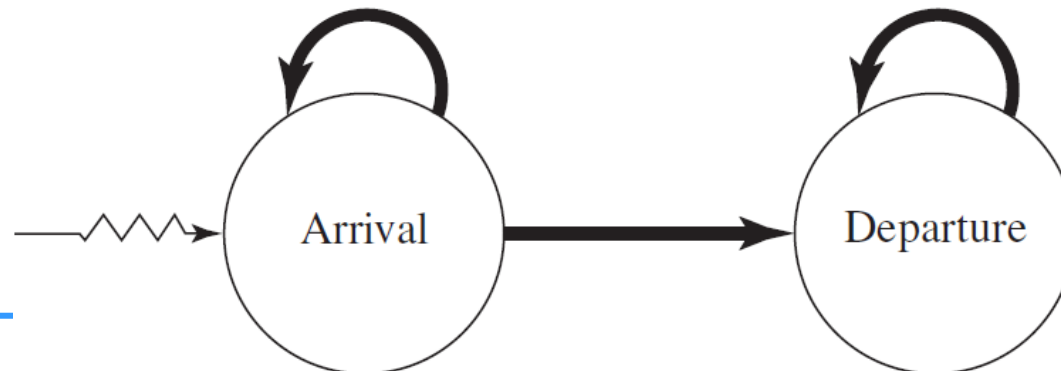
Muhammad Mahbub Alam, PhD

Professor

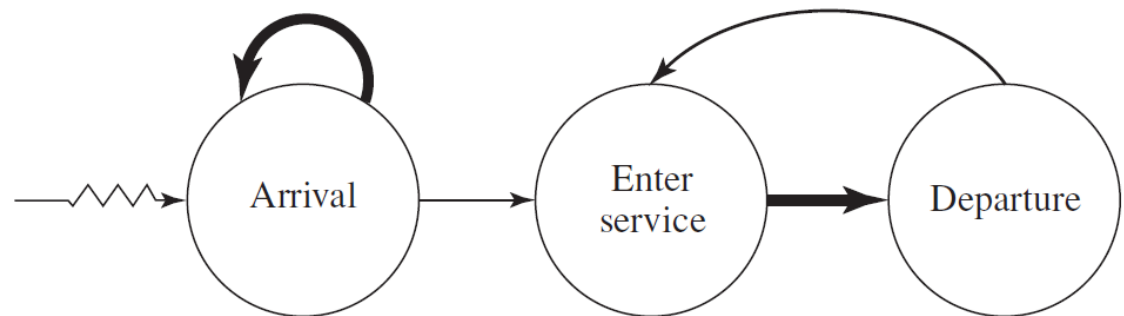
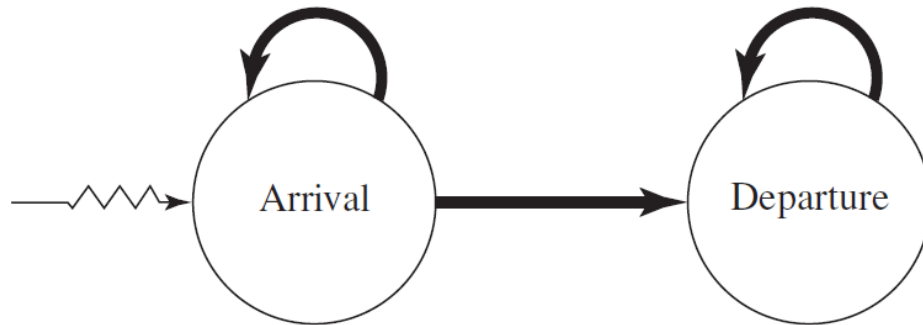
CSE Department, IUT

Event Graph

- Proposed events, each represented by a *node*, are connected by *directed arcs*
- Depicts how events may be scheduled from other events and from themselves
 - the arrival event schedules another future occurrence of itself
 - and (possibly) a departure event
 - departure event may schedule another future occurrence of itself
 - the arrival event must be initially scheduled in order to get the simulation going
- the heavy, smooth arrows indicate that an event at the end of the arrow *may* be scheduled from the event at the beginning of the arrow in a (possibly) *nonzero* amount of time
- thin jagged arrow indicates that the event at its end is scheduled initially
- the arrival event reschedules itself and may schedule a departure
- the departure event may reschedule itself

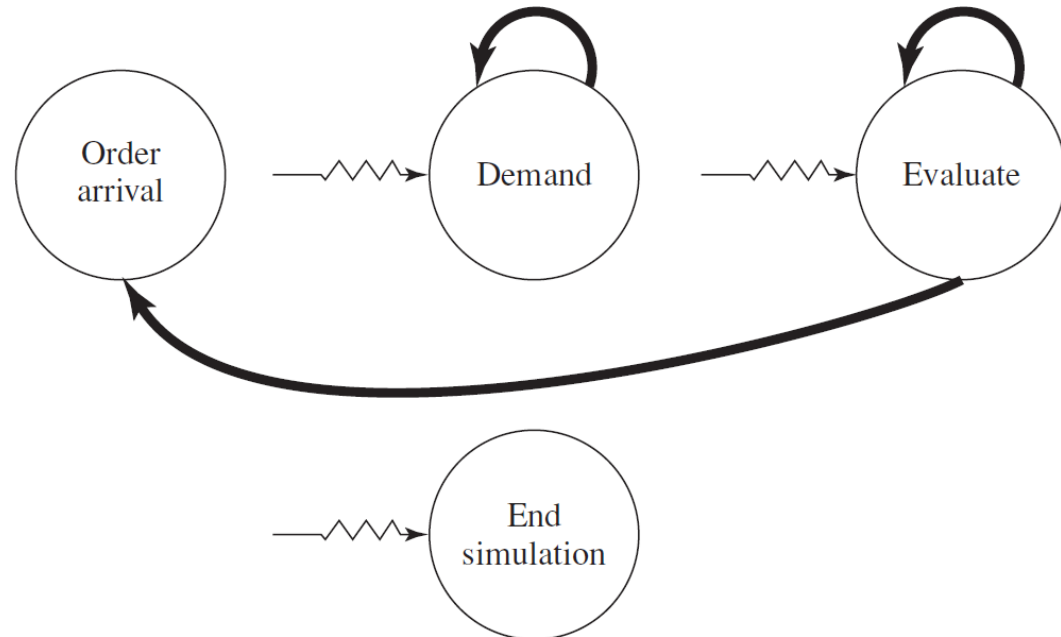
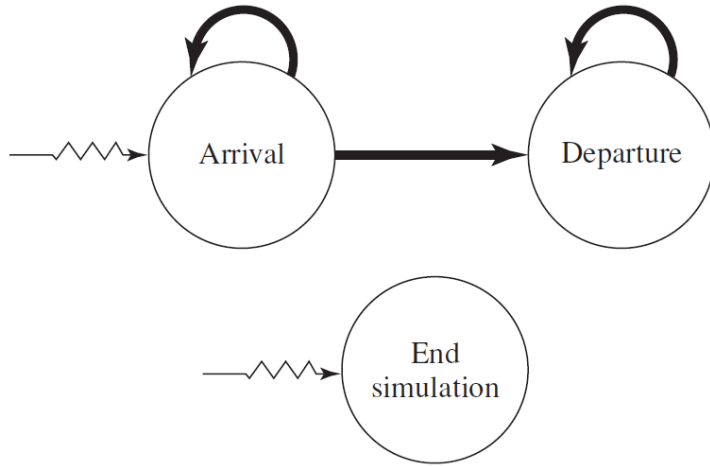


Event Graph



- The two thin smooth arrows each represent an event at the beginning of an arrow potentially scheduling an event at the end of the arrow without any intervening time, i.e., immediately

Event Graph



Time Shared Computer Model

Problem Statement

- Operator thinks for a random time $\sim \text{expo}$ (25sec)
- Then sends a job to the CPU
 - A single queue (round robin)
- Job service time (s) $\sim \text{expo}$ (0.8 sec)
- CPU allocates a maximum of $q = 0.1$ sec
 - If $s > q \rightarrow$ allocate q
 - Else allocate s
- Job swap time $\sim$ deterministic (0.015 sec)
- Unfinished job, after each run, is returned to the queue
- Finished job is returned to the terminal
 - User start thinking

Conceptual Model

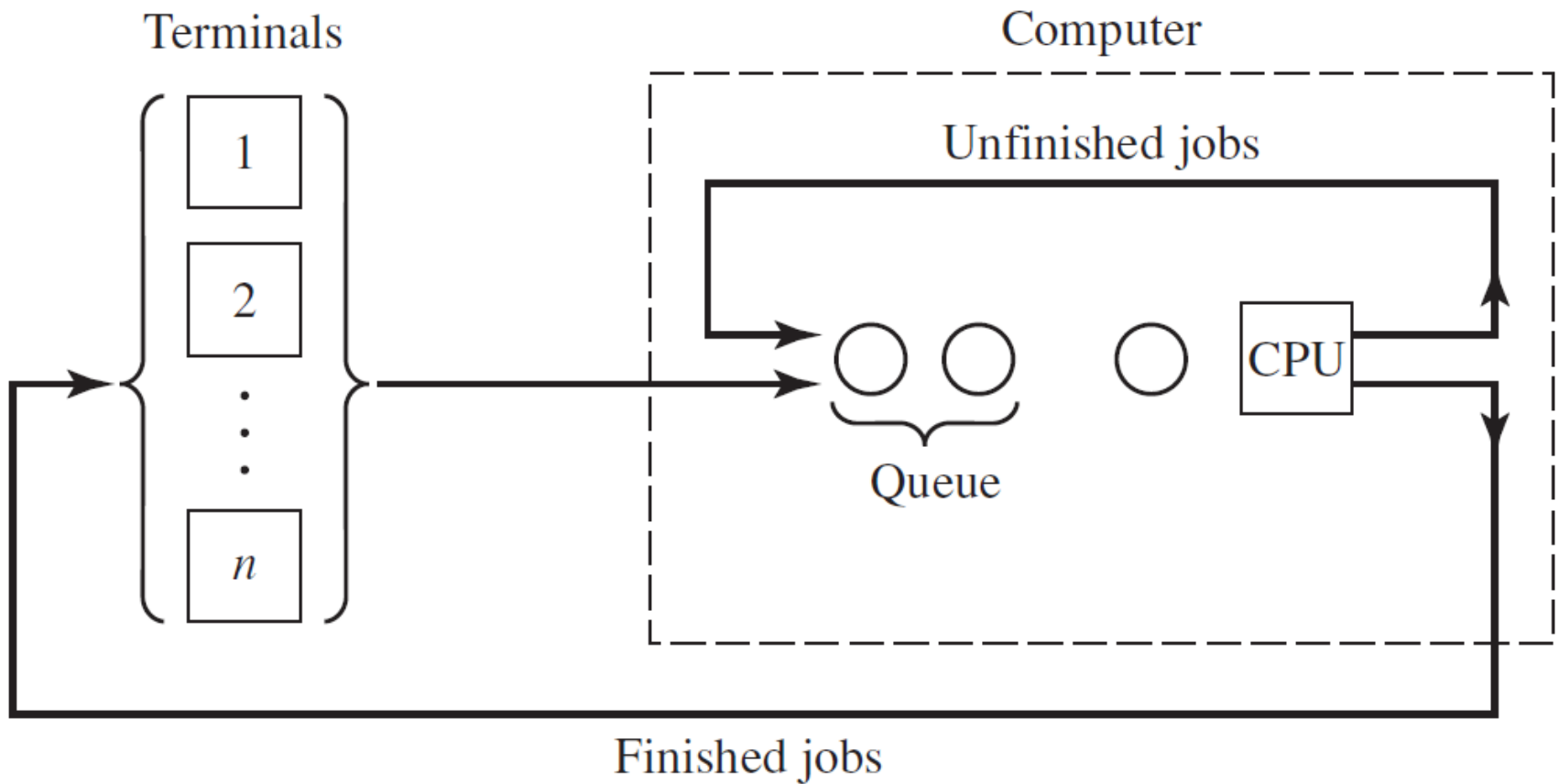


FIGURE 2.13

Time-shared computer model.

Event description	Event type
Arrival of a job to the CPU from a terminal, at the end of a think time	1
End of a CPU run, when a job either completes its service requirement or has received the maximum processing quantum q	2
End of the simulation	3

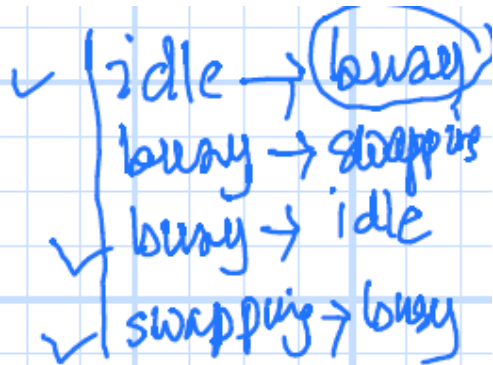
Conceptual Model

Events

- o arrival (1)
- o end-cpu-cycle (2)
- o end-swapping (3)

State variables

- o queue-length, $q(t)$
- o server-status, $x(t)$



$S = \{ (idle, 0), (busy, 0), (busy, 1), (swapping, 0), (swapping, 1), \dots \}$

$q(t) = \{0, 1, \dots, n\}$
 $x(t) = \{idle, busy, swapping\}$
 $= \{0, 1, 0\}$

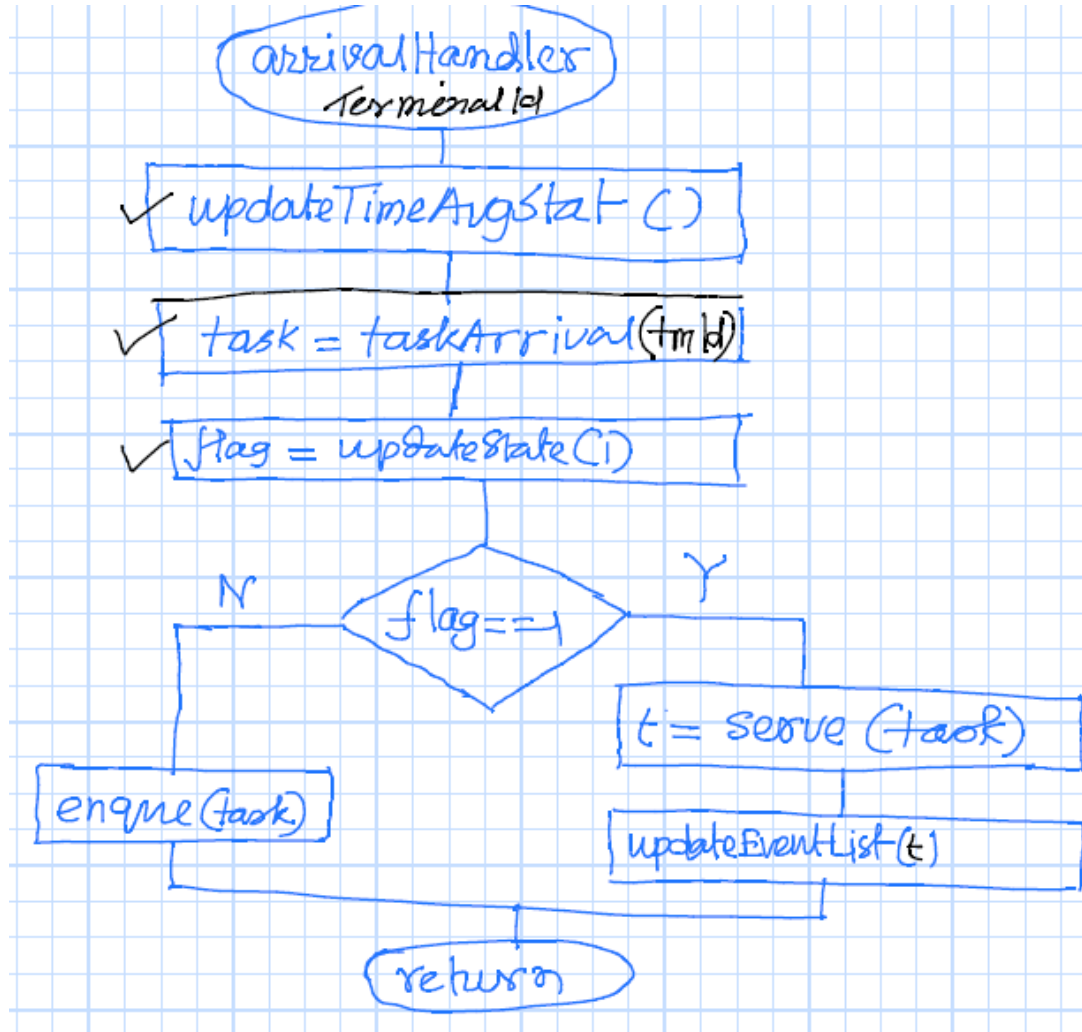
- Input and Output Variables:
- Average inter-arrival time rate, time quanta, swap time, average service time
- Average delay, utilization, time-average # of job in the queue, server

Specification Model

$$q(t) = \begin{cases} q(t) + 1, & \text{arrival, } x(t) \neq \text{idle} \\ q(t) - 1, & \text{end-cpu-cycle,} \\ & \text{rem-time} = 0 \\ q(t), & 0/w. \end{cases}$$

$$x(t) = \begin{cases} \text{busy,} & \text{if arrival, } x(t) = \text{idle} \\ & \text{if end-swapping} \\ \text{idle,} & \text{if end-cpu-cycle, remtime} = 0, \\ & q(t) = 0. \quad (\underline{x(t) = \text{busy}}) \\ \text{swapping,} & \text{if end-cpu-cycle,} \\ & \text{rem-time} > 0 \parallel q(t) > 0 \\ \underline{x(t)}, & \underline{0/w} \end{cases}$$

Computational Model



updateTimeugStat

InterEventTime = now() - timeLastEvent

areaJobTimeQue += InterEventTime

areaBusyTimeServer += InterEventTime

return

updateState
eventId

eventId == 1

x(t) == idle

x(t) = busy

return 0

eventId == 2

q(t) == 0

x(t) = idle

return 2

q(t) --

return 3

